LEF MODEL · CHAPTER 1

Chapter 1: Darwin's Problem and the Corotation Revolution

Based on Paper 02 — LEF Mechanism in Principle (Introduction and sections A.1–A.3)

01 · THE MYSTERY

The mystery: how do you leave the Earth without being thrown off it?

In the Prologue we met the Moon's eight great mysteries — and the first hint of the key that opens all of them together: the Moon was built from the Earth's own material while still attached to it. But this is exactly where the sharpest physical question arises, the one that brought down the idea of an Earth origin for more than a hundred years.

Suppose for a moment that you are standing on the equator of the young Earth, about 4.5 billion years ago. This early Earth is glowing hot, swollen with heat, and spinning at a dizzying rate: it completes a full turn about its axis every **5.5 hours** — four times faster than today! And yet, if you try to release a lump of rock or a drop of magma from the height of this Earth's surface, it will not soar into orbit around the Earth. It will drop straight back to the ground like a stone thrown from the top of a tower.

Why? Because even with a fast 5.5-hour rotation, the Earth's surface does not move fast enough to overcome gravity. In 1879 the British astronomer and mathematician **George Darwin** (the son of Charles Darwin) proposed that the young Earth spun so fast that it split and cast the Moon out of itself. But within a few decades physicists showed that if the Moon had broken away directly from the surface, the Earth would have had to spin at an impossible rate of less than

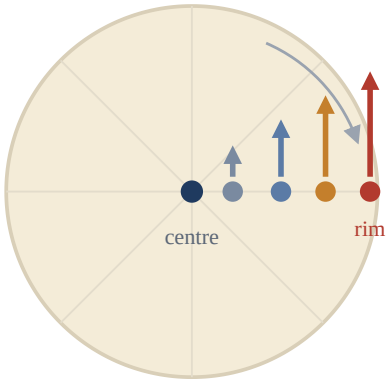
three hours per day — a state requiring almost three times (and up to four times in the early calculations) more angular momentum than everything that exists in the Earth–Moon system today.

> The historical verdict seemed sealed: a moon cannot be taken out of the Earth.

But Darwin and his successors made one hidden assumption that nobody challenged — **that the material must break away from the surface itself**. What happens if the material is not thrown from the surface, but rises gradually upward while remaining attached to the Earth and turning together with it?

02 · AN IMAGE

An image: the girl who walks from the centre of the merry-go-round to its rim



Same rotation rate, a bigger circle — **a higher speed**

Imagine a large merry-go-round in an amusement park turning at a steady, constant rate — say, one full turn every ten seconds. Nobody pushes it harder, and its motor does not speed up. Now imagine a girl standing near the centre of the merry-go-round who begins to walk slowly, step by step, across the turning floor toward the outer rim.

With every step she takes away from the centre, something remarkable happens without anyone throwing her: the circle she completes every ten seconds becomes larger and larger. To complete a wider circle in exactly the same ten seconds, **her linear speed must grow**. Near the central axis she barely moves;

halfway out the wind already strikes her face; and when she reaches the outer rim she is flying at tremendous speed — even though the rotation rate of the merry-go-round itself has not changed by a fraction of a second!

Corotation

This is exactly the principle of **corotation** (turning together): every part attached to the rotating Earth completes one circuit every Earth day, so the higher it rises above the surface, the larger the sideways speed that the Earth's own rotation gives it — while the gravity trying to pull it down actually grows weaker.

03 · SCIENCE AND MODEL

Science and model: two questions that must be kept apart

To seriously test any theory in which the Moon was born out of the Earth, Paper 02 makes clear that we must separate **two entirely separate mechanical questions** that must not be mixed:

QUESTION 1

The release-speed question (the kinematic test)

At the moment the material is released, is its sideways (tangential) speed large enough to keep it in a free orbit around the Earth without falling back to the surface?

QUESTION 2

The angular-momentum-budget question (the conservation test)

Assuming we have found a point where the release speed matches an orbit — did the early Earth have enough angular momentum in the first place to bring the material to that point, and still be left with the angular momentum we measure in today's Earth–Moon system?

In Darwin's classical fission theory, these two questions were tied together in the wrong place — at the Earth's surface — and so both failed. In the LEF model, by

contrast, we first set out the exact environmental conditions of the early Earth (called here **the "parent body"**), and see how both questions receive a simple, clean answer.

An important mass clarification · no double counting

Before the Moon separated, all the material it is made of was inside the young Earth. Therefore the mass of the **initial parent body** (M_{tot}) is the sum of the mass of today's Earth ($M_{\oplus} = 5.972 \times 10^{24}$ kg) and of today's Moon ($m = 7.35 \times 10^{22}$ kg) together: 6.0455×10^{24} **kg**. No mass disappears and none is counted twice: as the Moon's mass moves outward, the Earth's mass falls correspondingly to 5.972×10^{24} kg.

In addition, the young Earth was hot throughout and therefore **thermally swollen** (an adopted expansion of about 120 km beyond the cold radius of 6,397 km), so that its volumetric radius stood at **6,517 km** and its mean density at about 5,214 kg/m³. Because of the fast rotation of 5.5 hours per day, the semi-fluid parent body took the shape of a **Maclaurin spheroid** (the equilibrium shape of a rotating fluid body — flattened at the poles and bulging at the equator), with an equatorial radius of about **6,721 km** and a polar radius of about **6,128 km**.

6.0455×10²⁴	6,517	6,721	6,128
Mass of the parent body (kg)	Volumetric radius (km)	Equatorial radius (km)	Polar radius (km)

04 · THE MECHANISM IN WORDS

The mechanism in words: closing the abyss and the bridge of the pear

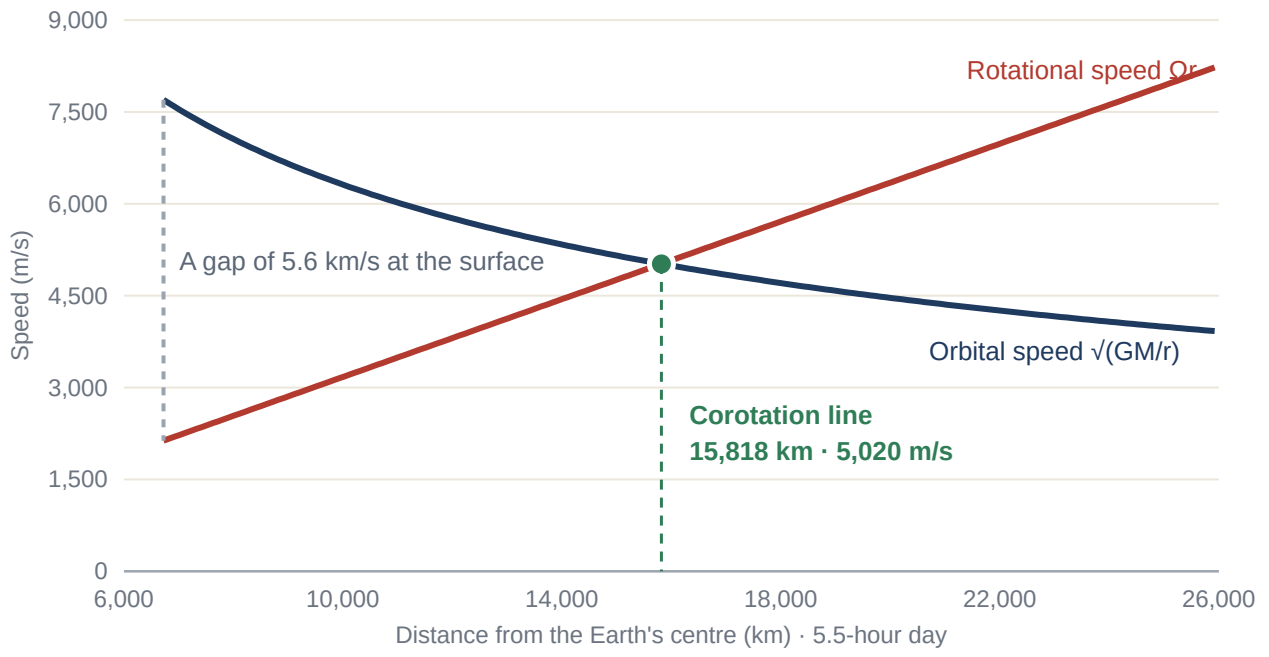
Now let us check what happens with real numbers on our parent body, which completes a full turn every 5.5 hours (19,800 seconds):

> **At the surface (the equator, 6,721 km from the centre):** the Earth's rotation gives a point on the surface a sideways speed of only **2,133 metres per**

second. By contrast, to stay in a free circular orbit at this height without falling requires a speed of **7,701 metres per second**. The gap — **5,568 metres per second** — is enormous. Material released here would enter a narrow elliptical orbit whose semi-major axis is only 3,495 km, that is, an orbit that plunges at once deep into the Earth.

On the way up (corotation): when the material stays attached to the Earth and climbs upward, the two speeds move toward each other: the speed given by rotation grows in a straight line with distance, while the speed needed for a free orbit decreases as gravity weakens.

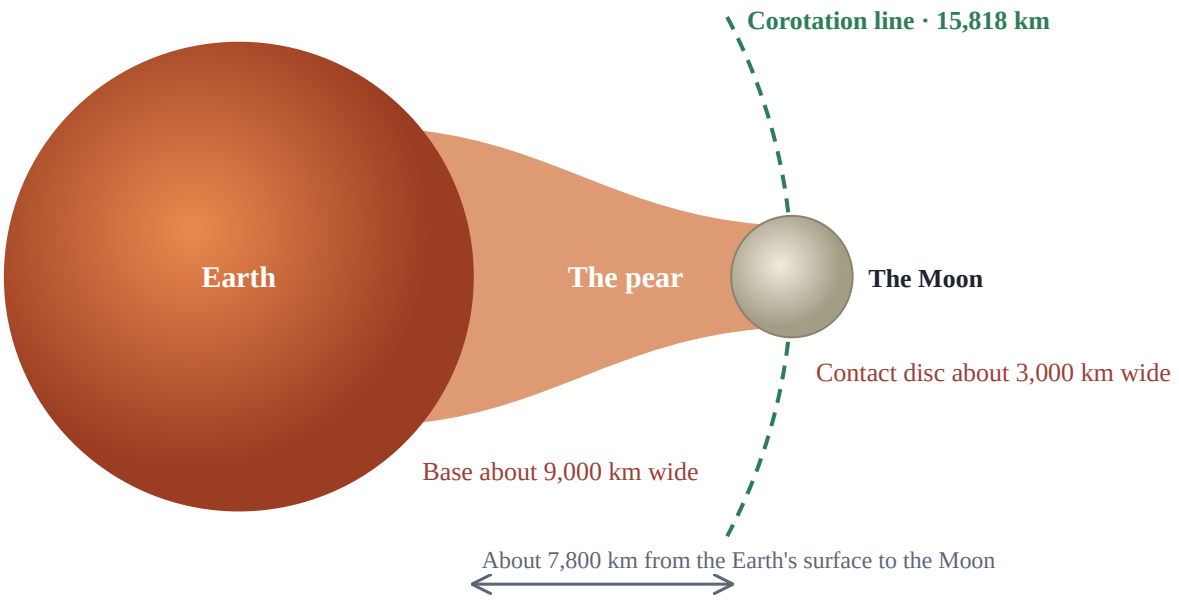
The meeting point — the corotation line (15,818 km): at a distance of **15,818 km** from the Earth's centre (about 2.4 times the Earth's radius), the two speeds meet at exactly **5,020 metres per second**. Here the gap disappears completely! A body brought here and released is not thrown — at the moment of detachment it is already moving at exactly the speed of a free circular orbit.



The two speeds as a function of distance from the Earth's centre, for a 5.5-hour day: the gap of 5,568 m/s at the surface closes at the corotation line.

> But how does the material reach such a height without falling apart on the way?

Here the central structure of the model enters the picture: **the "pear"**. This is not a thin pipe or a volcano spraying lava into space, but an enormous, continuous hill made of the Earth's own material: its base on the Earth's surface is remarkably wide — about **9,000 km** (and over the full range 9,000–11,000 km), its sides curve in a smooth arc with no narrow "waist" that could break, and its upper tip touches the forming Moon across a contact disc about **3,000 km** wide.



The Earth, the pear and the Moon to scale: the whole pear lies inside the corotation line, with the Moon at its upper tip. (Schematic drawing; dimensions as in the text.)

This pear bridges a distance of about **7,800 km** between the Earth's surface and the Moon, and it lies **entirely inside the corotation line**. The physical meaning is enormous: at every point along the pear, the Earth's gravity is stronger than the centrifugal push of the rotation. So the pear is held firmly to the Earth as an inseparable part of it and does not fly off; only the body sitting at its upper tip reaches the corotation line and is released as a whole unit!

05 · THE KEY EQUATION

Academic explanation and the key equation

To find the corotation radius (R_{sync}), we equate the tangential speed of a body turning together with the Earth at angular velocity $\omega = 2\pi/T$ with the speed required for a free circular orbit in a central gravitational field:

The speed given by rotation	$v_{\text{rot}} = \omega R$
The speed for a free circular orbit	$v_{\text{circ}} = \sqrt{\dfrac{GM}{R}}$
The point of equality	$\omega R_{\text{sync}} = \sqrt{\dfrac{GM}{R_{\text{sync}}}}$
The corotation radius	$R_{\text{sync}} = \left(\dfrac{GM}{\omega^2}\right)^{1/3} = \left(\dfrac{GMT^2}{4\pi^2}\right)^{1/3}$

This formula teaches us a simple basic law: the corotation radius depends only on the mass of the central body (M) and on the length of its day (T) to the power of two-thirds ($T^{2/3}$). The faster the Earth spins (a shorter day), the closer the corotation line comes to it; the more its rotation slows (a longer day), the farther out the corotation line moves.

Why 5.5 hours?

And why is 5.5 hours the natural starting point?

Because if we take all the angular momentum of today's Earth and Moon ($3.441 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}}$) and return it into a young, swollen ($R_{V,0} = 6{,}517$ km) and relatively uniform ($k = 0.4$) Earth, we get a day of **5.54 hours**. Since the solar tide has removed about 1% of the system's angular momentum over the 4.5 billion years that have passed since then, the true starting point was slightly faster — a day of **5.50 hours** ($L_0 \simeq 3.466 \times 10^{34}\ \mathrm{kg\,m^2\,s^{-1}}$).

06 · IN THE SKY TODAY

The signature in reality: the geostationary orbit above our heads

The corotation formula is not a theoretical exercise on a blackboard — it is at work right now above our heads and makes modern global communication possible. Communication and weather satellites (geostationary satellites) are launched exactly to **the corotation radius of today's Earth**.

With a 24-hour day (exactly 23 hours and 56 minutes of sidereal rotation), this line lies at a height of about **35,786 km** above the surface (about 42,164 km from the Earth's centre). Any satellite there completes one orbit in exactly the time the Earth completes one rotation, so it "hovers" permanently above the same point on the equator. 4.5 billion years ago, when the day lasted only 5.5 hours, the same magic line lay much closer — at a distance of 15,818 km from the Earth's centre!



The corotation line then and now, on the same scale (distance from the Earth's centre).

The lab box: closing the speed gap and the starting conditions (Paper 02, Part A)

Parameter / test	At the surface (the equator)	At the initial corotation line	Physical meaning
Length of the day T_0	5.5 hours (19,800 s)	5.5 hours (19,800 s)	Rotation ~4 times faster than today (derived from conservation of angular momentum)
Distance from the Earth's centre	6,721 km (equatorial radius)	**15,818 km** (~2.4 Earth radii)	The pear bridges about 7,800 km above the surface
Rotational speed v_{rot}	2,133 m/s	**5,020 m/s**	Grows linearly with distance from the axis (ωr)
Circular orbital speed v_{circ}	7,701 m/s	**5,020 m/s**	Falls with distance as gravity weakens ($\sqrt{GM/r}$)
The speed gap Δv	**−5,568 m/s (falls back)**	**0 m/s (free orbit at once!)**	**The speed gap that brought down Darwin's model is closed**

07 · NEXT

The door ahead

We have solved the first riddle: when the release point is moved from the surface to the corotation line, the speed gap disappears completely. But the moment the pear begins to rise and carry mass outward, a new and fascinating question arises: **doesn't the rising of the pear slow the Earth's rotation and drive the corotation line too far away?**

In the next chapter — **Chapter 2: The Battle over the Length of the Day and the Angular-Momentum Balance** (Paper 02, sections A.4–A.8 and Part B) — we will discover how two opposing processes fight over the spin rate of the early Earth, and see how the angular-momentum budget closes to within 1.007.

Chapter 2

The Battle over the Length of the Day and the Angular-Momentum Balance →

← Back to the Prologue

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